JP Lacsamana

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Education

University of Oklahoma — Norman, OK
*B.S. Computer Science – **4.0*** — *Expected Graduation: Dec 2026*

Experience

Microsoft — Seattle, WA
Software Engineering Intern — *May 2026 – Aug 2026*

- Built a storage-agnostic crawler as an **Azure App Service** WebJob to detect, archive, and delete stale data
- Reduced storage costs 90% in testing — an estimated $20K savings over 5 years
- Built a competing-consumer **distributed crawler system** with distributed-locking states and key-space splitting
- Parallelized crawling with concurrent threading across the system, achieving a 256x speed-up
- Leveraged **C#/.NET** inheritance to make the crawler configurable and extensible across environments
- Enabled rapid deployment to production and org-wide adoption via custom filters for other teams

Google — Pittsburgh, PA
Software Engineering Intern (ASDI) — *May 2025 – Aug 2025*

- Spearheaded development of "AI Overview" into Google's Programmable Search Engine (PSE)
- Shipped as the default layout for the "AI Overview" component
- Built the Gemini-powered feature end-to-end (**Java**, **JavaScript**, **HTML/CSS**) on GCP
- Enabled PSE clients to embed Gemini-augmented search directly into their own sites
- Improved "AI Overview" accuracy and recency using **RAG** to ground Gemini responses in live search results
- Designed the "AI Overview" interface with a UX designer, researching a comparable feature's interaction patterns

Momentus Technologies — Remote
Software Development Intern — *May 2024 – Aug 2025*

- Migrated 30+ production Vue 2 components to Vue 3 across the "Momentus Elite" platform
- Validated features against **RavenDB-backed** test fixtures, shipping with zero production incidents
- Containerized dev builds with **Docker** and deployed via **AWS**

University of Oklahoma — Norman, OK
William Kerber Scholar Teaching Assistant — *Aug 2026 – Present*

- Selected as 1 of 3 Kerber Scholars to tutor 20+ students over 3 semesters
- Covered core CS concepts (OOP, data structures, algorithms, theory of computation) and **Python/Java**
- Whiteboarded problems into scaffolded subproblems, researching advanced topics to close gaps

Projects

Beach Box | Realtime Boxing Web Game Using Computer Vision - *Mediapipe, Supabase, Vercel* — Apr 2026

- **Winner at Hook'em Hacks** among 50+ teams at UT Austin's hackathon
- Built the game's punch-detection system using webcam-based **MediaPipe** hand tracking
- Estimated punch speed from the rate of change in apparent fist size, thresholding to register a hit
- Cross-validated hits using pinky knuckle-line rotation as a secondary signal to reduce false positives
- Synced real-time damage state between players using **Supabase** and deployed on **Vercel**
- Shipped a playable multiplayer game with a 4-person team in under 24 hours

Autonomous Robot Tour Guide | Software for Turtlebot4 - *Ros2, Python, sklearn, Nav2* — May 2026

- Designed and developed TurtleBot 4 tour-guide software on **ROS 2**
- Built a controller node in **Python** to manage tour state from sensor input
- Built a **DBSCAN**-based person-detection system with **sklearn**, clustering LiDAR returns into leg pairs
- Determined whether a person/group was still following, triggering wait/resume tour logic
- Built the waypoint navigation system using **Nav2** and LiDAR-based SLAM
- Enabled the robot to navigate accurately to input waypoints across the real-world robotics lab floor plan

Leadership and Technical Skills

Languages: Python, Javascript, Java, C++, C, SQL, Typescript, HTML, CSS
Technologies: Git, React, Vue, Supabase, React Native, Azure, Google Cloud, AWS, Vercel, Docker, Distributed Systems
Leadership Positions: Sigma Nu Director of Academic Affairs (1 year), Engineering Pathways Student Mentor (1 year)